HERIOT-WATT UNIVERSITY

F79MB Statistical Models B **Assessment Project 2 – 2024**

Your report for this project should be submitted through Turnitin by **3:30pm (local time at your campus) on Friday 5th April 2024**. A link to the submission page is available through 'Assessed Project 2 (60%)' in Modules or 'Assignments' on the course Canvas page. Please use the submission link provided to submit your work.

This project will account for **60%** of your grade for the course.

The total length of your report should be no more than 12 pages (11-point font, A4 size), including graphs and tables but excluding any appendices. Project title and student identification information should be in page 1, not a separate title page, and counts towards the 12-page limit. Please note that the **overall exposition/presentation is worth 10 marks** for this project.

You may consult with your lecturer or colleagues but your report must be your own work. **Plagiarism** is a serious academic offence and carries a range of penalties, some very serious – see Appendix A. You are required to submit a Standard Declaration of Authorship form. The form is available through 'Assessed Project 2' on the course Canvas page. **You must sign this Declaration of Authorship for Project 2 before you can have access to the project 2 questions**.

- Answer all tasks in this project, making sure that you add appropriate context and that your answers have a clear and logical structure and are well presented.
- You should explain carefully your work for each task so that your work demonstrates understanding of the methodology and computations that you use.
- You should include clearly labelled and correctly referenced graphs and add appropriate comments.
- You should use `R` to perform the required analyses and produce suitable graphs. Unless otherwise stated you do not need to explain the `R` commands that you use. However you must put the `R` code in appropriate appendices at the end of your report, and should include appropriate commenting within your `R` code.
- See Appendix B for the marks allocated to overall exposition/presentation.
- **Late project submissions will be penalised according to the University Policy on Submission of Coursework. That is, work submitted after the deadline but within 5 working days will be subject to a 30% deduction from the mark awarded; work submitted more than 5 working days after the deadline will be awarded a mark of zero. No individual extensions are permitted. In the case where a student submits coursework up to five working days late and has valid mitigating circumstances, the mitigating circumstances policy will apply.**

You can expect to receive feedback on your work by 6th May 2024.

Tasks:

1. A retail store is interested in understanding the factors influencing customer purchase behavior. They have collected data on several variables for a sample of customers, including:

   - `Purchase Amount`: The amount spent by the customer in dollars.
   - `Age`: The age of the customer.
   - `Membership Level`: A categorical variable indicating the customer's membership level (Silver, Gold, Platinum).

   The data in the `DataCustomer.csv` file, available in the '`Assessed Project 2 (60%)`' folder on Canvas. You can load the data into `R` using a command similar to `read.table("C:/R/DataCustomer.csv", header=T)`.

   Note that you will need to specify your own path to the file.

   (a) Present appropriate numerical and graphical summaries of the distributions of the three variables separately, and comment on your results.

   [4 marks]

   (b) Plot purchase amount against age, distinguishing between customers with different membership levels. Comment on your plot.

   [3 marks]

   (c) The company is interested in understanding the factors influencing purchase amount. Perform a multiple linear regression analysis with purchase amount as the response variable, age as a covariate, and membership level as a factor. Fit a model with main effects and an interaction term. Use appropriate analyses to decide which terms should be retained in the model. Your analysis should include R output, plots, comments, model checking, comprehensive conclusions, and discussions on the adequacy of the preferred fitted model.

   [14 marks]

   (d) Use your preferred fitted model to estimate the expected purchase amount of

      i. a 30-year-old customer with Platinum membership;
      ii. a 60-year-old customer with Gold membership.

   Then, based on the quantitative analysis, propose appropriate marketing strategies for the retail store's manager.

   [3 marks]

2. The dataset in the `icu_data.txt` file, available in the 'Assessed Project 2 (60%)' folder on Canvas, shows part of the data on the ICU study data set. It consists of a sample of 200 subjects who were part of a much larger study on survival of patients following admission to an adult intensive care unit (ICU). The primary objective of this study was to construct a logistic regression model aimed at forecasting the likelihood of survival until hospital discharge for these patients, considering their medical conditions.

   You can load the data into `R` using a command like `read.table("C:/R/icu_data.txt", header=T)`. Note that you will need to specify your own path to the file. The variables given are:

   `AGE` = age of the patient (years)

   `MC` = medical condition (arthritis, asthma, cancer, diabetes, hypertension, obesity)

   `STA` = vital status at hospital discharge (Lived/Death)(Lived = 0, Death = 1)

   Note that both `MC` and `STA` are categorical variables.

   It is proposed to model the data in a generalized linear model framework, modelling `STA` as the response variable with `AGE` and `MC` as explanatory variables.

   (a) Present appropriate numerical and graphical summaries of the distributions of the 3 variables separately, and comments on your results.

   [4 marks]

   (b) Produce an appropriate plot to explore the relationship between the response variable `STA` and the explanatory variable `AGE`, and comment.

   [2 marks]

   (c) Carry out an appropriate test of independence to investigate if the vital status at hospital discharge (`STA`) is dependent on medical condition (`MC`). Your report should include a contingency table, a p-value and a clear conclusion.

   [3 marks]

   (d) Write an appropriate generalized linear model with `STA` as the response variable and `AGE` and `MC` as explanatory variables (without an interaction term). Clearly define the terms used in the model. Then, discuss why a logistic model is more suitable than a linear regression model.

   [4 marks]

   (e) Analyze the data by fitting an appropriate generalized linear model discussed in (d). Carry out appropriate analyses to determine which terms should be retained in the model. Your analysis should include appropriate R output, plots, comments, model checking, comprehensive conclusions, and discussions on potential enhancements to the fitted model.

   [13 marks]

[Overall exposition/presentation: 10 marks]

[Project total: 60 marks]

[END OF PROJECT]

Appendix A: Plagiarism

- Coursework reports must be written in a student's own words and any code in their coursework must be their own code. If some text or code in the coursework has been taken from other sources, these sources must be properly referenced.
- Failure to reference work that has been obtained from other sources or to copy the words and/or code of another student is plagiarism and if detected, this will be reported to the School's Discipline Committee. If a student is found guilty of plagiarism, the penalty could involve voiding the course.
- Students must never give hard or soft (electronic) copies of their coursework reports or code to another student. Students must always refuse any request from another student for a copy of their report and/or code.
- Sharing a coursework report and/or code with another student is collusion, and if detected, this will be reported to the School's Discipline Committee. If found guilty of collusion, the penalty could involve voiding the course.
- See `https://www.hw.ac.uk/students/studies/examinations/plagiarism.htm` for more details.

Appendix B: Rubric for marks allocated to overall exposition/presentation

The 10 marks available for the overall exposition/presentation of your report will be awarded according to the scale below.

3 marks	• Some answers lack clarity and logical structure • Some analyses may not be relevant • Statements of conclusions are present but may not be very clear • Some statistical calculations and methodology set out for the reader • Some tables and figures relevant and referenced
5 marks	• Answers sometimes clearly and logically structured • Focus on key points but may include some superfluous/irrelevant analyses • Statements of conclusions generally suitable for a non-statistician • Statistical calculations and methodology are generally presented okay for the reader • Tables and figures mostly well chosen, clear, and correctly referenced • Some sources are clearly and correctly referenced, with R code included
8 marks	• Answers are mostly clear and logically structured • Clear focus on key points, avoiding superfluous/irrelevant analyses • Statements of conclusions suitable (wherever possible) for a non-statistician • Statistical calculations and methodology set out clearly for the reader • Tables and figures well chosen, clear, correctly referenced and easy to interpret • Sources used clearly and correctly referenced • R code included with clear comments
10 marks	• As in the previous category, but the answers are very clear, strong, and logically structured. Additionally, the work demonstrates mathematical sophistication and insight, with evidence of original thought and reasoning. It also maintains a highly focused approach on key points, without superfluous or irrelevant analyses. R code is included with clear comments and without superfluous coding.